

Joshua Alvarez

126 Bucks Avenue
Cape May Court House, NJ 08210
📞 (609) 602-9615
✉️ Joshuaalvarez13972@gmail.com

Professional Profile

Full-stack and AI/ML software engineer specializing in clean code, scalable architecture, and intelligent systems. Experienced in building, deploying, and optimizing robust web applications and machine learning models from concept to production.

Technical Skills

Languages	Python, Java, JavaScript, TypeScript, C++, HTML, CSS, Bash, SQL
AI & ML	PyTorch, TensorFlow, OpenAI SDK, Generative AI, GANs, Computer Vision
Frameworks	React, Node.js, Flask, Git, Unicorn, Nginx, Hibernate
Cloud & DB	AWS EC2, Docker, PostgreSQL, MySQL, H2

Software Engineering Projects

- 2024–2025 **Progressive Growing GAN (ProGAN)**, *Machine Learning Research*
- Implemented a Progressive Growing GAN with Wasserstein loss (WGAN-GP) for high-fidelity image synthesis.
 - Achieved best FID score of 10.93, trained on NVIDIA RTX 4090 GPU using PyTorch.
 - Applied optimization techniques relevant to simulation rendering and synthetic data generation pipelines.
- 2024–2025 **Self-Paced AI Learning Platform**, *REU Grant Recipient*
- Designed and implemented a self-paced learning platform using Flask, OpenAI API, and dynamic quiz generation.
 - Built a modular architecture for dynamic subjects and learning progression tracking.
 - Deployed scalable web application to AWS EC2 using Unicorn and Nginx.
- 2023–2024 **Express Auto Web Application**, *Full-Stack Development*
- Developed full-stack web application utilizing Flask backend, React (JSX) frontend, and PostgreSQL database.
 - Integrated robust third-party APIs including Twilio SMS messaging and SendGrid email support.
 - Designed and optimized responsive front-end for mobile and desktop platforms.

Professional Experience

- May 2025–Present **IT Manager**, *The Reeds at Shelter Haven*, Stone Harbor, NJ
- Manage high-availability infrastructure and internal systems for a luxury resort environment.
 - Develop and deploy automated scripts for network infrastructure, security systems, and VLAN monitoring.
 - Actively streamline technical workflows and troubleshoot deployed software environments.
- Jan 2026–May 2026 **Systems Engineer (Contract)**, *Cloud Speak*
- Programmed and configured an HTML5 cloud-based hospitality content management platform.
 - Developed multi-site network integrations ensuring low-latency communication across 3 distinct properties.
 - Troubleshoot complex backend system issues to maintain continuous operational availability.

Education

- Dec 2025 **Bachelor of Science in Computer Science**, *Stockton University*, Galloway, NJ, GPA: 3.86
- **Honors:** Research Experience for Undergraduates (REU) Recipient for AI Research.
 - **Relevant Coursework:** Data Structures & Algorithms I & II, Software Engineering, Machine Learning.